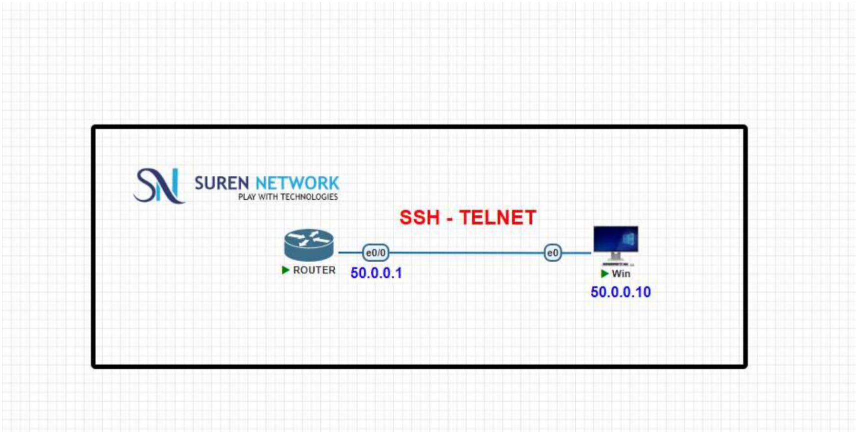
CCNA Lab Report

Router Configuration & Introduction to Telnet



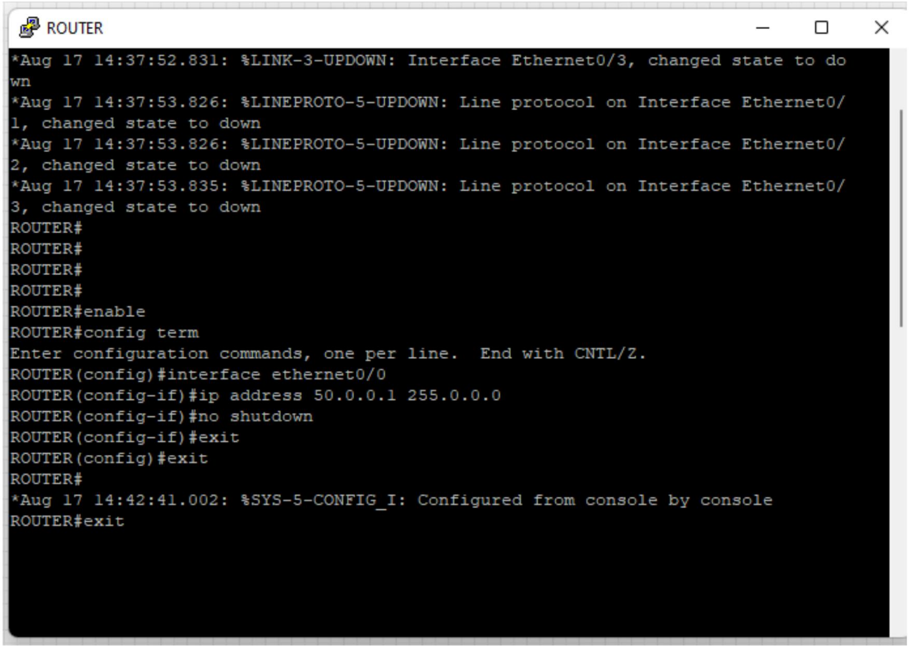
Lab Topology — Router (50.0.0.1/8) ↔ Windows PC (50.0.0.10/8)

Objective	Configure a router's Ethernet interface, verify PC-router connectivity, and set up local/console authentication as groundwork for Telnet access.
Tool	Cisco IOS Router + Windows PC (EVE-NG / Packet Tracer lab)
Topic	Router Configuration and Intro to Telnet – CCNA

Step 1 | Configure the Router Interface

The router boots with all interfaces down. Ethernet0/0 is assigned an IP address in the 50.0.0.0/8 network and brought up with **no shutdown**.

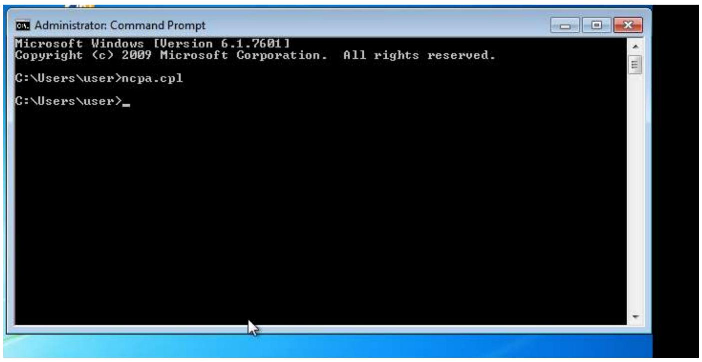
```
ROUTER#enable
ROUTER#config term
ROUTER(config)#interface ethernet0/0
ROUTER(config-if)#ip address 50.0.0.1 255.0.0.0
ROUTER(config-if)#no shutdown
ROUTER(config-if)#exit
ROUTER(config)#exit
```



Router console: link-down messages on unused interfaces, followed by Ethernet0/0 configuration.

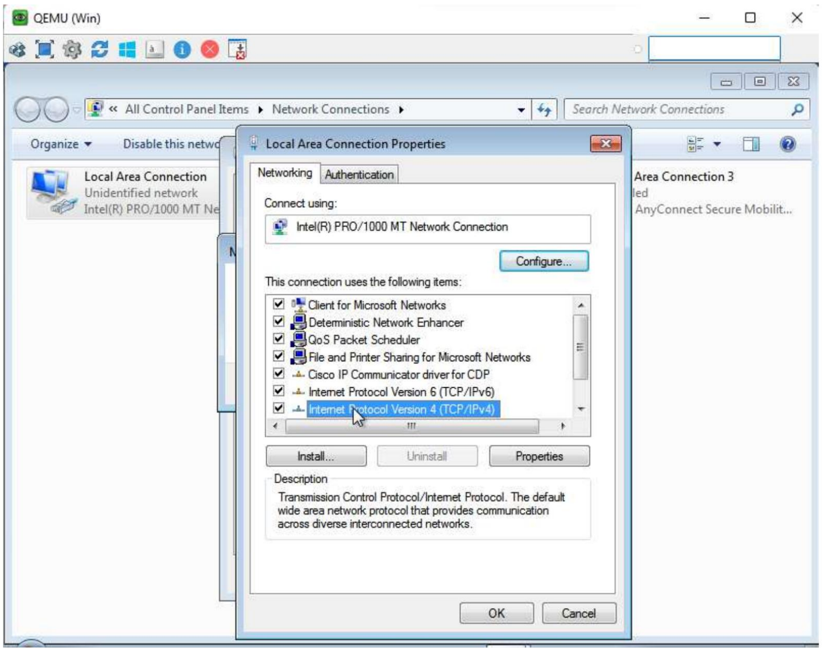
Step 2 | Configure the Windows PC IP Address

On the Windows PC, **ncpa.cpl** is run from Command Prompt to open Network Connections directly.

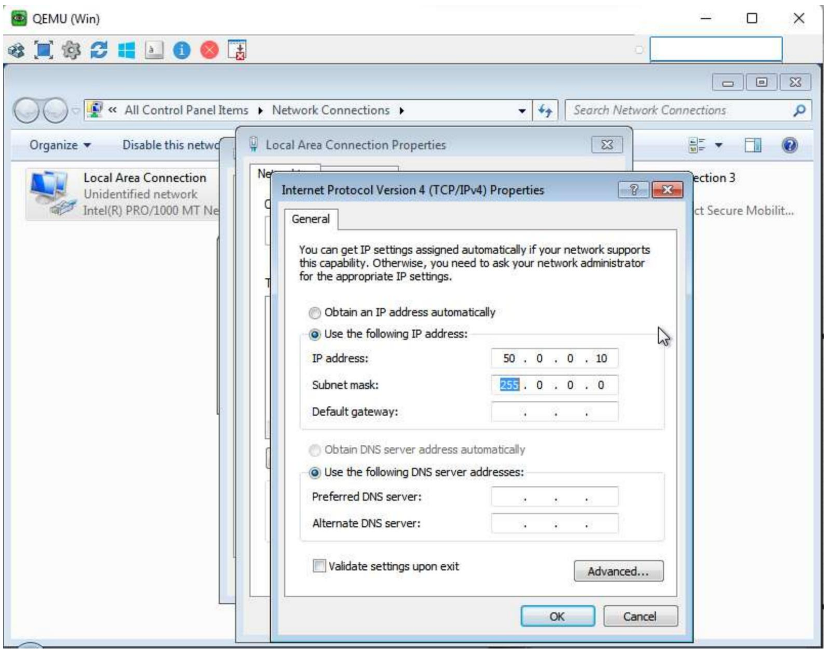


Running ncpa.cpl to jump straight to Network Connections.

The Local Area Connection's **IPv4 Properties** are opened to set a static address.



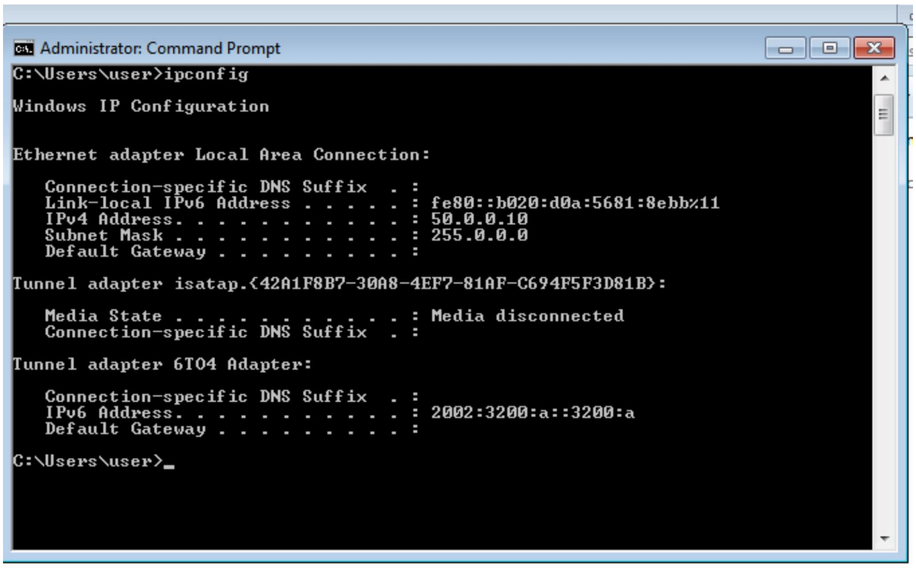
Network Connections → Local Area Connection Properties → TCP/IPv4.



Static IP configured: 50.0.0.10 / 255.0.0.0 (no gateway needed — same subnet).

Verified with **ipconfig**:

```
C:\Users\user>ipconfig

Ethernet adapter Local Area Connection:
IPv4 Address. . . . . . . . . . . : 50.0.0.10
Subnet Mask . . . . . . . . . . . : 255.0.0.0
```

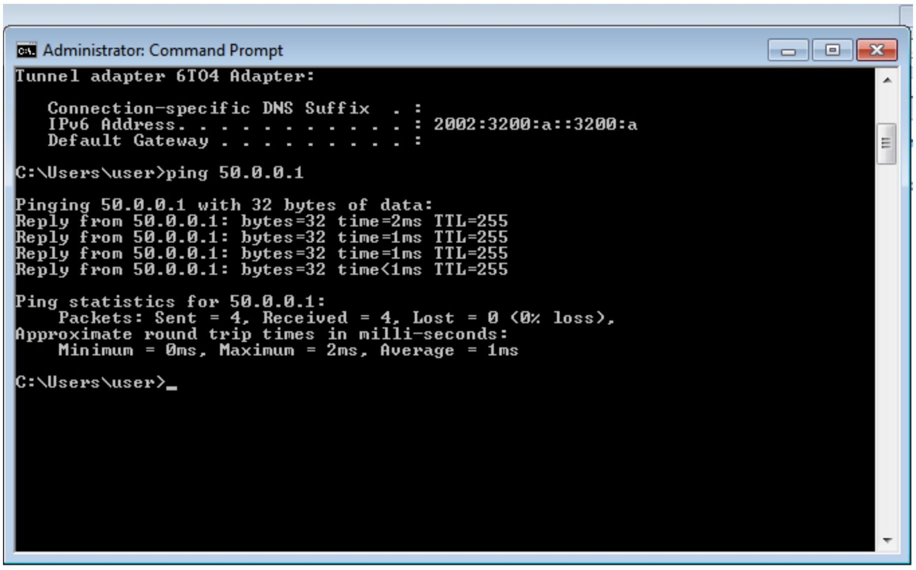


ipconfig output confirming the assigned IPv4 address.

Step 3 | Verify Connectivity

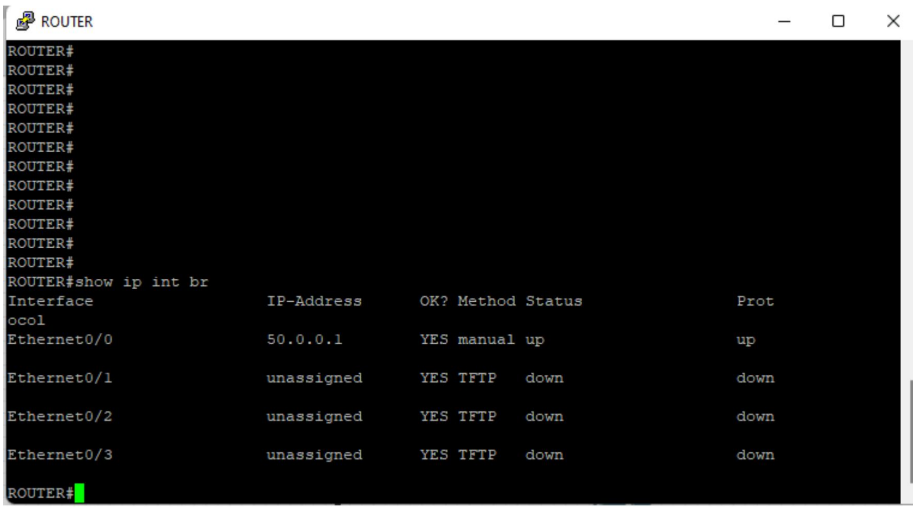
A ping from the PC to the router confirms Layer 3 reachability:

```
C:\Users\user>ping 50.0.0.1

Reply from 50.0.0.1: bytes=32 time=2ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time=1ms TTL=255
Reply from 50.0.0.1: bytes=32 time&lt;1ms TTL=255

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```



Successful ping — 0% packet loss between PC and router.

Confirmed on the router side with **show ip int br**:

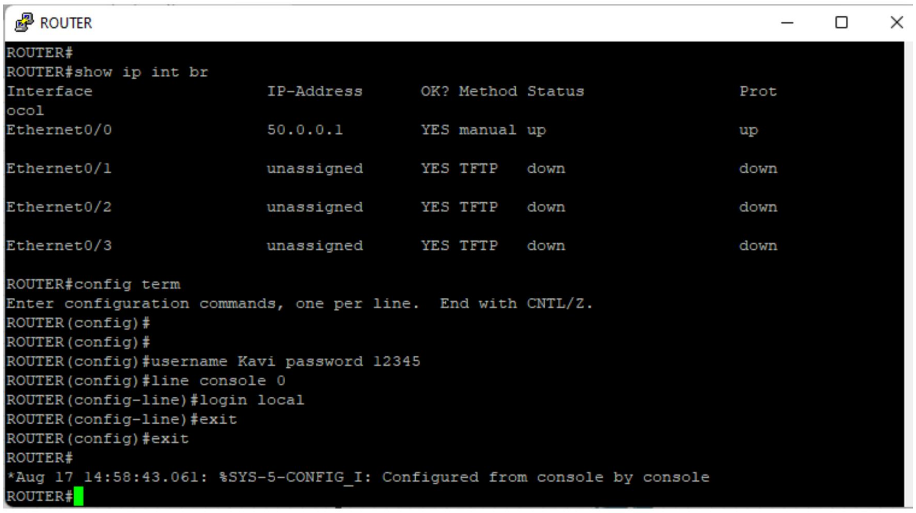


Ethernet0/0 shows up/up with IP 50.0.0.1; unused interfaces remain down.

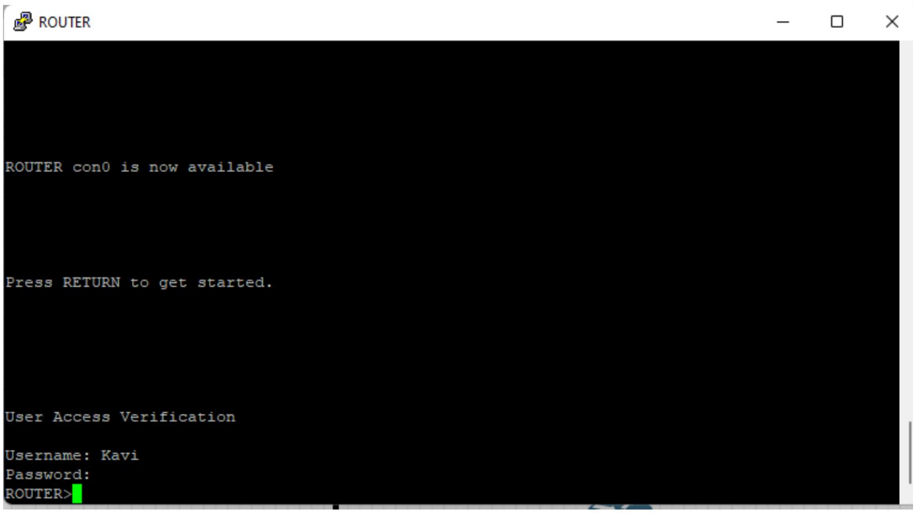
Step 4 | Local Username & Console Login

A local user account is created and the console line is set to authenticate against it using **login local**.

```
ROUTER(config)#username Kavi password 12345
ROUTER(config)#line console 0
ROUTER(config-line)#login local
ROUTER(config-line)#exit
ROUTER(config)#exit
```



Local username 'Kavi' created and console line set to login local.

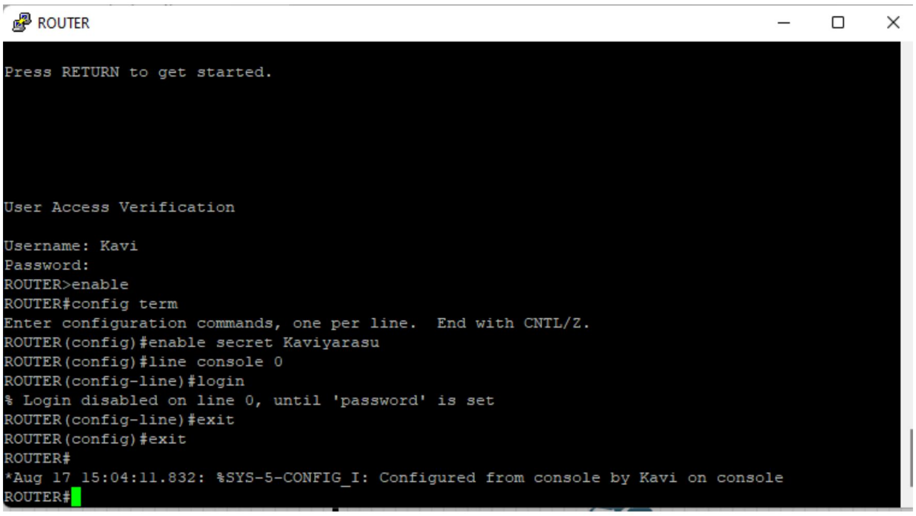


Console re-prompts for the local username/password before granting access.

Step 5 | Enable Secret Password

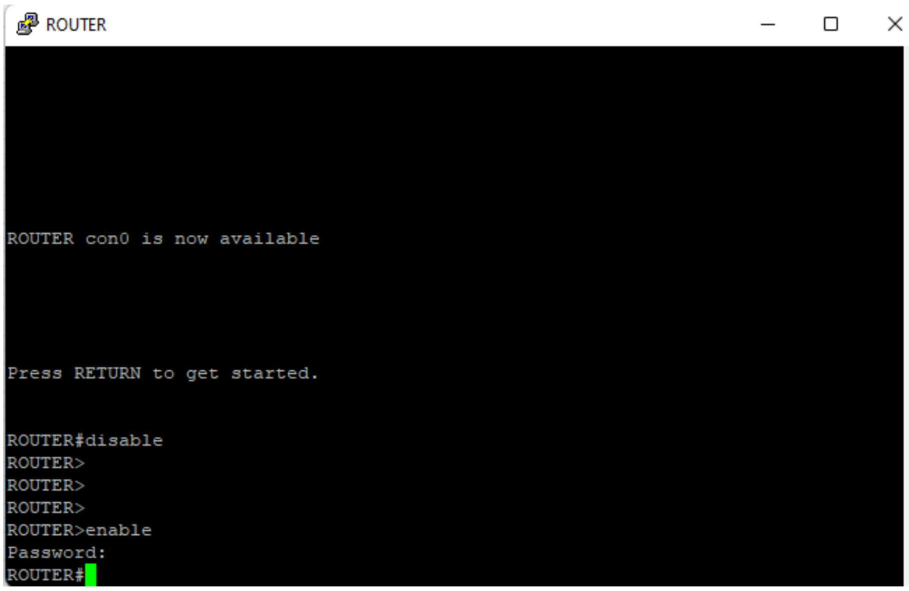
An **enable secret** (encrypted) password is set to protect privileged EXEC mode.

```
ROUTER(config)#enable secret Kaviyarasu
```



enable secret configured; entering enable now prompts for this password.

Note: attempting **login** (without **local**) on *line console 0* without a line password first produced the expected IOS message: `% Login disabled on line 0, until 'password' is set`. This is normal — plain **login** requires a **password** to be set on that line, unlike **login local** which uses the username database instead.

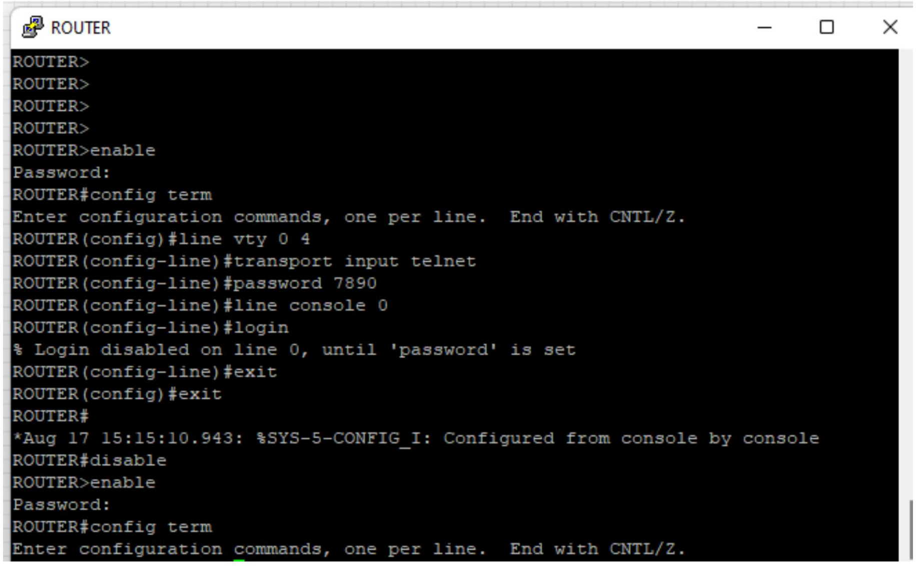


Testing privileged mode: disable → enable now asks for the enable password.

Step 6 | Prepare VTY Lines for Telnet

Finally, the VTY (virtual terminal) lines are configured to accept incoming Telnet connections with a password, laying the groundwork for remote access in the next stage of the lab.

```
ROUTER(config)#line vty 0 4
ROUTER(config-line)#transport input telnet
ROUTER(config-line)#password 7890
```



VTY lines 0-4 configured for Telnet with password 7890.

Key Learnings

- **no shutdown** is required to bring an interface administratively up.
- Both ends of a link need matching subnet/mask for Layer 3 connectivity.
- **ping** and **show ip int br** are the go-to tools to verify reachability.
- **username** + **login local** enables local-database authentication on a line.
- **enable secret** stores an encrypted privileged-mode password (preferred over **enable password**).
- **line vty 0 4** + **transport input telnet** + **password** is the minimum setup for Telnet remote access.

Lab documented as part of CCNA practice — Router Configuration & Intro to Telnet.